

**Backpaper - Linear Algebra I (2025-26)**

**Time: 3 hours.**

*Attempt all questions.*

1. Prove or disprove: If  $B$  is a basis of  $V$  and  $S$  is a subspace of  $V$  then  $B$  contains a basis of  $S$ . [5 marks]
2. Show that a subset  $A$  of a vector space is linearly dependent if and only if there exists  $\mathbf{x} \in A$  such that  $\mathbf{x} \in \text{Sp}(A - \{\mathbf{x}\})$ . [5 marks]
3. Let  $V$  be the vector space of all functions from  $\mathbf{R}$  to  $\mathbf{R}$ . Let  $V_e$  be the subset of even functions:  $f(-x) = f(x)$  for all  $x$ , and let  $V_o$  be the subset of odd functions:  $f(-x) = -f(x)$  for all  $x$ .
  - (a) Prove that  $V_e$  and  $V_o$  are subspaces of  $V$  [3 marks]
  - (b) Prove that  $V_e + V_o = V$ .  
(Hint: for a function  $f$  consider the functions  $[f(x) + f(-x)]/2$  and  $[f(x) - f(-x)]/2$ ) [4 marks]
  - (c) Prove that  $V_e \cap V_o = \{0\}$  [3 marks]
4. Let  $T : V \rightarrow W$  be a linear transformation. Show that  $T$  is 1-1 if and only if the null space of  $T$  is  $\{0\}$ . [5 marks]
5. Let  $T$  be the linear operator on  $\mathbf{R}^3$  defined by

$$T(x_1, x_2, x_3) = (3x_1 + x_3, -2x_1 + x_2, -x_1 + 2x_2 + 4x_3).$$

What is the matrix of  $T$  in the ordered basis  $\{\mathbf{x}, \mathbf{y}, \mathbf{z}\}$ , where  $\mathbf{x} = (1, 0, 1)$ ,  $\mathbf{y} = (-1, 2, 1)$  and  $\mathbf{z} = (2, 1, 1)$ ? [5 marks]

6. Obtain a rank-factorization of

$$\begin{bmatrix} 5 & 1 & 3 \\ 0 & 0 & 2 \\ 10 & 2 & 4 \end{bmatrix} \quad [5 \text{ marks}]$$

7. Let  $\mathbf{A}$  be an  $m \times n$  matrix over  $F$ . Show that  $\mathbf{A}$  has a right inverse if and only if  $\mathcal{C}(\mathbf{A}) = F^m$ . [5 marks]
8. Solve the system  $\mathbf{Ax} = \mathbf{b}$  (find whether it is consistent and a general solution if it is consistent). Here

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} \quad [5 \text{ marks}]$$

9. In the vector space  $\mathbf{R}^4$ , find two different complements of the subspace  $S = \{(x_1, x_2, x_3, x_4) : x_3 - x_4 = 0\}$ . [5 marks]